



Aim To Top Review and Training Center - Iligan
Agricultural and Biosystems Engineering Licensure Examination
Review Session

MASTERING AGRICULTURAL PROCESSING

100 CONCEPT-BASED QUESTIONS FOR RICE AND CORN

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INTRODUCTION

Agricultural and Bio-Processing refers to local activity or series of activities to maintain or raise the quality or change the form or characteristics of agricultural, fishery, forestry and biological products/materials and includes, but is not limited to, cleaning, sorting, grading, treating, drying, dehydrating, grinding, mixing, milling, canning, dressing, slaughtering, freezing, pasteurizing, conditioning, packaging, repacking, transporting of agricultural, fishery, forestry and other biological products/materials

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QUESTION 1

The purposeful movement of air at a low rate through a product to maintain or improve product quality.

- a. Ventilation
- b. Aeration
- c. Cooling
- d. Airing

ANSWER



b. Aeration

Explanation: Aeration is a low-rate airflow process used to maintain or improve the quality of stored grains.

- a. Ventilation refers to general air circulation in a space, not specific to product preservation.
- c. Cooling is a result of airflow but not the purpose.
- d. Airing is a non-technical term and not standard in drying or storage.

QUESTION 2

What is the primary purpose of drying harvested grains like rice and corn?

- a. To improve the color and aroma
- b. To reduce weight for transportation
- c. To lower the moisture content and prevent spoilage
- d. To separate the grain from impurities

ANSWER



c. To lower the moisture content and prevent spoilage

Explanation: Drying reduces grain moisture to safe levels for storage and prevents spoilage due to microbial growth.

- a. Color and aroma are secondary effects.
- b. While drying reduces weight, it's not the primary reason.
- d. Cleaning handles impurity removal, not drying.

QUESTION 3

Which of the following is most affected when paddy is dried at high temperatures above 50°C?

- a. Milling recovery
- b. Germination rate
- c. Foreign matter content
- d. Moisture content

ANSWER



b. Germination rate

Explanation: High drying temperatures damage the embryo, reducing seed viability.

- a. Milling recovery may be affected, but less critically.
- c. Drying doesn't affect foreign matter; cleaning does.
- d. Moisture content is intended to be reduced, not damaged.

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QUESTION 4

The separation of kernels from cobs in corn is referred to as:

- a. Milling
- b. Shelling
- c. Winnowing
- d. Cleaning

ANSWER



b. Shelling

Explanation: Shelling removes kernels from the cob.

- a. Milling refers to grain processing.
- c. Winnowing separates light impurities.
- d. Cleaning removes foreign particles.

QUESTION 5

What postharvest operation involves removing foreign matter such as stones, straw, and chaff from rice grains?

- a. Grading
- b. Shelling
- c. Drying
- d. Cleaning

ANSWER



d. Cleaning

Explanation: Cleaning removes physical contaminants from grains.

- a. Grading separates based on size or quality.
- b. Shelling applies to corn, not rice.
- c. Drying is for moisture removal.

QUESTION 6

In grain drying, what is the period where moisture is primarily removed from the grain surface, similar to evaporation from a water surface?

- a. Absorption phase
- b. Equilibrium phase
- c. Constant rate period
- d. Falling rate period

ANSWER



c. Constant rate period

Explanation: This is when free moisture is removed as in open evaporation.

- a. Absorption involves moisture uptake.
- b. Equilibrium refers to balance with the environment.
- d. Falling rate involves internal moisture movement.

QUESTION 7

Which type of dryer has the drying air and the grain moving in opposite directions?

- a. Cross-flow dryer
- b. Concurrent-flow dryer
- c. Counter-flow dryer
- d. Tunnel dryer

ANSWER



c. Counter-flow dryer

Explanation: Air and grain move in opposite directions in a counter-flow dryer.

- a. Cross-flow: air moves perpendicularly.
- b. Concurrent: same direction.
- d. Tunnel: batch conveyed through tunnel.

QUESTION 8

Which of the following dryers causes the product to be in suspension by the drying air?

- a. Tunnel dryer
- b. Indirect-fired dryer
- c. Fluidized or spouted bed dryer
- d. Cross-flow dryer

ANSWER



c. Fluidized or spouted bed dryer

Explanation: Suspension by airflow is a feature of fluidized bed dryers.

a, b, d: these operate with grounded grains.

QUESTION 9

What is the most common reason to avoid using high temperatures when drying seeds intended for planting?

- a. Color darkening
- b. Reduced germination viability
- c. Faster drying rate
- d. Incomplete husk removal

ANSWER



b. Reduced germination viability

Explanation: High temp can damage embryos and lower seed viability.

a: Visual, not functional impact.

c: True, but at a cost.

d: Husk removal is not drying's primary job.

QUESTION 10

What does the 'heat utilization factor' (HUF) indicate in drying operations?

- a. Fuel efficiency per hour
- b. Ratio of ambient to drying air temperature
- c. Efficiency of drying air temperature use
- d. Time taken to reach equilibrium moisture

ANSWER



c. Efficiency of drying air temperature use

Explanation: $HUF = (T1 - T3) / (T2 - T1)$ indicates how well heat is used in drying.

a: Related to combustion, not HUF.

b: Not a ratio used in HUF.

d: Unrelated to heat utilization.

QUESTION 11

Which of the following is used to measure grain moisture content?

- a. Barometer
- b. Hygrometer
- c. Moisture meter
- d. Psychrometer

ANSWER



c. Moisture meter

Explanation: Moisture meters are specifically designed to measure the moisture content of grains.

a: Barometers measure atmospheric pressure.

b: Hygrometers measure humidity in the air.

d: Psychrometers also relate to air humidity, not grain moisture.

QUESTION 12

What happens during the falling rate period in drying?

- a. Moisture is evaporated from the surface only
- b. Grain temperature stays constant
- c. Internal moisture diffuses to the surface
- d. The rate of drying is constant

ANSWER



c. Internal moisture diffuses to the surface

Explanation: This phase involves the movement of moisture from the grain interior to its surface.

a: That's the constant rate period.

b: Temp rises in falling rate.

d: The drying rate decreases.

QUESTION 13

What grain damage is caused by too rapid drying at high temperatures?

- a. Mold growth
- b. Stress cracks
- c. Browning
- d. Fungal spots

ANSWER



b. Stress cracks

Explanation: Rapid drying leads to internal stress and cracking.

a, c, d: More associated with storage or spoilage.

QUESTION 14

Which grain component contains the highest oil content?

- a. Bran
- b. Germ
- c. Endosperm
- d. Husk

ANSWER



b. Germ

Explanation: The germ is rich in oils and nutrients.

a: Bran contains fiber.

c: Endosperm is starch-rich.

d: Husk is mostly fiber and protective material.

QUESTION 15

What part of the corn kernel remains after producing grits?

- a. Germ
- b. Pericarp
- c. Endosperm
- d. Tip cap

ANSWER



c. Endosperm

Explanation: Corn grits are made from the starchy endosperm.

a, b, d: Removed in grit production.

QUESTION 16

What is the term for the natural drying of grains under the sun?

- a. Mechanical drying
- b. Controlled drying
- c. Solar drying
- d. Field drying

ANSWER



d. Field drying

Explanation: Field drying occurs naturally post-harvest in the open.

a, b: Involve equipment.

c: Refers to solar-assisted drying, not open field.

QUESTION 17

What type of energy is primarily used in mechanical dryers?

- a. Electrical
- b. Solar
- c. Biomass
- d. Thermal

ANSWER



d. Thermal

Explanation: Dryers rely on thermal energy (from fuel) to remove moisture.

a: Powers fans and controls.

b: Limited application.

c: Sometimes used as thermal source, but not the main category.

QUESTION 18

Which dryer type involves a fixed grain bed and a continuous flow of heated air?

- a. Cross-flow dryer
- b. Batch dryer
- c. Circulating dryer
- d. Silo dryer

ANSWER



a. Cross-flow dryer

Explanation: Air flows perpendicular to grain bed in cross-flow dryers.

b: Not continuous.

c: Mixes grain during drying.

d: Used for storage/drying but not with cross-airflow.

QUESTION 19

Which grain component primarily contains protein?

- a. Bran
- b. Germ
- c. Husk
- d. Endosperm

ANSWER



d. Endosperm

Explanation: Endosperm contains both starch and significant protein.

a, b: Also nutritious but not the main source in flour.

c: Low nutritional content.

QUESTION 20

Which of the following is a key parameter in determining dryer capacity?

- a. Grain color
- b. Air pressure
- c. Moisture reduction per hour
- d. Grain type

ANSWER



c. Moisture reduction per hour

Explanation: Dryer capacity is measured in terms of moisture removal rate.

a: Not relevant.

b: Affects airflow, not capacity alone.

d: Affects drying time but not direct capacity metric.

QUESTION 21

Which postharvest practice is used to remove lightweight impurities by air?

- a. Winnowing
- b. Shelling
- c. Gritting
- d. Milling

ANSWER



a. Winnowing

Explanation: Winnowing separates lighter materials using airflow.

b: Removes kernels from cobs.

c: Produces corn grits.

d: Grain processing step.

QUESTION 22

Which device in drying systems regulates temperature automatically?

- a. Relay
- b. Thermostat
- c. Breaker
- d. Timer

ANSWER



b. Thermostat

Explanation: Thermostats sense and control temperature.

a, c, d: Electrical components not specific to temp control.

QUESTION 23

What term describes damage from grain absorbing moisture from air during storage?

- a. Sweating
- b. Cracking
- c. Rewetting
- d. Sprouting

ANSWER



c. Rewetting

Explanation: Grains reabsorb moisture when stored in humid air.

a: Refers to moisture condensation.

b: From drying stress.

d: Due to moisture + warmth.

QUESTION 24

What is the ideal final moisture content for safe storage of corn?

- a. 14%
- b. 12%
- c. 18%
- d. 10%

ANSWER



a. 14%

Explanation: Corn is safe for storage at or below 14% moisture.

b: Too dry, may be unnecessary.

c: Too high for safe storage.

d: Also safe but not standard.

QUESTION 25

Which process in rice milling removes the outer husk of the grain?

- a. Polishing
- b. Hulling
- c. Whitening
- d. Sieving

ANSWER



b. Hulling

Explanation: Hulling removes husk; a key step before whitening.

a: Removes bran.

c: Removes more bran after hulling.

d: Separates by size.

QUESTION 26

Which type of corn has a dent at the broad end caused by shrinkage of soft starch?

- a. Flint corn
- b. Popcorn
- c. Sweet corn
- d. Dent corn

ANSWER



d. Dent corn

Explanation: Dent corn has soft starch at the center that shrinks on drying, forming a dent.

a: Flint corn has hard starch and no dent.

b: Popcorn is for popping, not identified by dent.

c: Sweet corn is eaten fresh; no dented kernel.

QUESTION 27

What postharvest operation converts paddy to rice by removing husk and bran?

- a. Cleaning
- b. Milling
- c. Sifting
- d. Conditioning

ANSWER



b. Milling

Explanation: Milling removes husk and bran to produce white rice.

a: Cleaning only removes foreign matter.

c: Sifting separates particles by size.

d: Conditioning adjusts moisture before further processing.

QUESTION 28

The volumetric measure of air movement, in cubic meters per minute, is referred to as:

- a. Static pressure
- b. Velocity
- c. CFM
- d. Volume flux

ANSWER



c. CFM

Explanation: CFM (cubic feet per minute) measures air volume flow.

a: Static pressure measures force, not volume.

b: Velocity is speed, not volume.

d: Volume flux is a less common term.

QUESTION 29

What is the grain classification if 80% of brown rice kernels are longer than 6.5 mm but shorter than 7.5 mm?

- a. Short
- b. Medium
- c. Long
- d. Extremely long

ANSWER



c. Long

Explanation: “Long” is defined as 6.5–7.49 mm in length.

a: Short is < 5.5 mm.

b: Medium is 5.5–6.49 mm.

d: Extremely long is ≥ 7.5 mm.

QUESTION 30

Which of the following is not considered foreign matter in palay?

- a. Straw
- b. Stones
- c. Immature grains
- d. Sand

ANSWER



c. Immature grains

Explanation: Immature grains are part of palay, not foreign matter.

a, b, d: These are all foreign materials.

QUESTION 31

A machine used to separate broken and whole grains by size and shape:

- a. Paddy separator
- b. Grain cleaner
- c. Grader
- d. Sheller

ANSWER



c. Grader

Explanation: Graders separate grain based on size.

a: Paddy separator distinguishes paddy from brown rice.

b: Cleaner removes debris.

d: Sheller detaches kernels from cob (corn).

QUESTION 32

Which is the correct description of “moisture content (wet basis)”?

- a. $\text{Water} \div \text{dry matter}$
- b. $\text{Water} \div \text{total weight (water + dry matter)}$
- c. $\text{Total weight} \div \text{dry matter}$
- d. $\text{Dry matter} \div \text{water}$

ANSWER



b. Water ÷ total weight (water + dry matter)

Explanation: Wet basis uses the total material weight as denominator.

a: This is dry basis.

c, d: Incorrect formulations.

QUESTION 33

Which dryer allows grain and air to move in the same direction?

- a. Cross-flow
- b. Counter-flow
- c. Concurrent-flow
- d. Tunnel

ANSWER



c. Concurrent-flow

Explanation: Concurrent-flow aligns both air and grain direction.

a: Cross-flow moves air perpendicular.

b: Counter-flow moves air opposite.

d: Tunnel refers to structure.

QUESTION 34

A friction-type whitening machine that uses opposing rubber rolls to remove husk:

- a. Kiskisan
- b. Cono-type
- c. Rubber roll
- d. Horizontal abrasive

ANSWER



c. Rubber roll

Explanation: Rubber roll dehusks using shearing action.

a: Kiskisan is a frictional dehuller but with metal cylinder.

b: Cono-type uses discs.

d: Abrasive machines whiten, not dehusk.

QUESTION 35

What is the bulk density of paddy?

- a. 476 kg/m^3
- b. 576 kg/m^3
- c. 486 kg/m^3
- d. 596 kg/m^3

ANSWER



b. 576 kg/m³

Explanation: This is the standard bulk density.

a, c, d: Not the correct specified value.

QUESTION 36

Which of the following grains is considered “brewer’s rice”?

- a. Grain longer than $6/8$
- b. Grain passing 1.4 mm round perforation sieve
- c. Grain between $3/8$ and $6/8$
- d. Whole grain

ANSWER



b. Grain passing 1.4 mm round perforation sieve

Explanation: Brewer's rice is the smallest, passing through 1.4 mm sieve.

a: That's head rice.

c: That's large broken.

d: Whole grain is unbroken.

QUESTION 37

Which component of the drying system is responsible for air distribution under pressure?

- a. Burner
- b. Fan
- c. Duct
- d. Plenum

ANSWER



d. Plenum

Explanation: Plenum holds and distributes pressurized air.

a: Burner heats air.

b: Fan moves air.

c: Duct carries air but doesn't pressurize.

QUESTION 38

What refers to the amount of heat in air available for evaporating water from grains?

- a. Heat potential
- b. Drying energy
- c. Available heat
- d. Specific heat

ANSWER



c. Available heat

Explanation: This term is used in drying theory.

a, b: Not standard terms.

d: Specific heat is a material property.

QUESTION 39

Which classification applies to a kernel with length $\geq$ 3.0 times its width?

- a. Bold
- b. Slender
- c. Round
- d. Thick

ANSWER



b. Slender

Explanation: $L/W \geq 3.0$ is slender.

a: Bold is between 2.0 and 3.0.

c: Round is < 2.0 .

d: "Thick" is not a standard term here.

QUESTION 40

What happens during the falling rate period of drying?

- a. Surface moisture evaporates
- b. Air condenses
- c. Internal moisture migrates to surface
- d. No moisture removal

ANSWER



c. Internal moisture migrates to surface

Explanation: Falling rate is governed by internal diffusion.

a: That's during constant rate.

b: Condensation is not part of drying.

d: Moisture removal continues, just slower.

QUESTION 41

What machine separates paddy and brown rice based on specific gravity and length?

- a. Tray-type separator
- b. Compartment-type separator
- c. Sieve separator
- d. Cono-type mill

ANSWER



a. Tray-type separator

Explanation: Tray-type uses gravity and length differences.

b: Uses buoyancy and gravity.

c: Not used for this purpose.

d: Used for dehusking.

QUESTION 42

What device reduces fuel pressure to a safe constant delivery pressure?

- a. Solenoid valve
- b. Orifice
- c. Pressure regulator
- d. Burner nozzle

ANSWER



c. Pressure regulator

Explanation: Regulates pressure despite upstream fluctuations.

a: Controls flow on/off.

b: Restricts flow, doesn't regulate pressure.

d: Delivers fuel, not regulates.

QUESTION 43

Which of the following is a disadvantage of bag storage?

- a. Easy to fumigate
- b. Requires mechanical handling
- c. Difficult to monitor insects
- d. Flexible stacking

ANSWER



c. Difficult to monitor insects

Explanation: Bag storage makes pest control hard.

a, b: Apply to bulk storage.

d: Bag storage is flexible.

QUESTION 44

What part of a friction whitener moves grains into the milling cylinder?

- a. Roller
- b. Worm feeder
- c. Screen
- d. Hopper

ANSWER



b. Worm feeder

Explanation: It forces grains into milling area.

a: Roller performs the whitening.

c: Screen separates bran.

d: Hopper holds grains.

QUESTION 45

Which is an advantage of bulk storage over bag storage?

- a. Lower capital cost
- b. More flexible
- c. Easier pest control and fumigation
- d. Easy to move grains

ANSWER



c. Easier pest control and fumigation

Explanation: Bulk storage allows better monitoring and control.

a & b: Apply to bag storage.

d: Both methods need handling equipment.

QUESTION 46

Which term refers to grain with $\frac{3}{8}$ to less than $\frac{6}{8}$ length of whole grain?

- a. Brewer's rice
- b. Head rice
- c. Small broken
- d. Large broken

ANSWER



d. Large broken

Explanation: This is the classification for this size.

a: Smaller than this.

b: Longer than $6/8$.

c: Smaller than $3/8$.

QUESTION 47

What happens to static pressure as airflow is forced through more grain depth?

- a. Increases
- b. Decreases
- c. Remains constant
- d. Oscillates

ANSWER



a. Increases

Explanation: *More depth creates more resistance to airflow.*

b: Only true if flow is reduced.

c, d: Not accurate.

QUESTION 48

Which burner operation cycles between high and low fire settings?

- a. Cycling burner
- b. Constant pilot
- c. Continuous burner
- d. Manual ignition

ANSWER



a. Cycling burner

Explanation: It cycles to control temperature.

b: Describes pilot only.

c: Doesn't cycle.

d: Not automated.

QUESTION 49

Which of the following provides the most energy-efficient drying?

- a. Indirect-fired dryer
- b. Tunnel dryer
- c. Direct-fired dryer
- d. Sun drying

ANSWER



c. Direct-fired dryer

Explanation: Combustion heat is directly used, reducing losses.

a: Heat transfer loss is higher.

b: May be batch or continuous.

d: Not consistent.

QUESTION 50

What refers to the drying air temperature minus ambient temperature?

- a. Airflow rate
- b. Drying efficiency
- c. Temperature rise
- d. Pressure drop

ANSWER



c. Temperature rise

Explanation: This is how much the air is heated.

a: Not temperature related.

b: This is a performance metric.

d: Refers to airflow resistance.

QUESTION 51

What milling machine uses centrifugal force to whiten rice?

- a. Cono-type mill
- b. Kiskisan mill
- c. Centrifugal type mill
- d. Rubber roll mill

ANSWER



c. Centrifugal type mill

Explanation: It relies on centrifugal force for milling.

a: Cono uses abrasive discs.

b: Kiskisan uses friction.

d: Rubber roll is used for husk removal, not whitening.

QUESTION 52

What is the function of dunnage in bag storage?

- a. Increase ventilation
- b. Prevent contact with floor moisture
- c. Increase bag stacking height
- d. Store fumigants

ANSWER



b. Prevent contact with floor moisture

Explanation: Dunnage keeps bags off the floor to prevent moisture absorption.

a: Ventilation is a side effect, not the purpose.

c: Dunnage doesn't increase stacking.

d: Not for fumigants.

QUESTION 53

The white or chalky portion on brittle grains is termed as:

- a. Broken grain
- b. Chalky kernel
- c. Yellow grain
- d. Germ

ANSWER



b. Chalky kernel

Explanation: Chalky kernels are brittle and white.

a: Refers to size, not appearance.

c: Yellow is due to heat or fermentation damage.

d: Germ is part of the seed, not damage

QUESTION 54

What postharvest practice equalizes grain moisture or temperature before further processing?

- a. Conditioning
- b. Aeration
- c. Tempering
- d. Polishing

ANSWER



c. Tempering

Explanation: Tempering ensures uniform moisture inside the grain.

a: Conditioning is preliminary treatment.

b: Aeration is for storage quality.

d: Polishing affects appearance.

QUESTION 55

What property of paddy is 1.11?

- a. Kernel density
- b. Bulk density
- c. Specific gravity
- d. Friction coefficient

ANSWER



c. Specific gravity

Explanation: The specific gravity of the kernel is 1.11.

a: Not labeled as density here.

b: Bulk density is 576 kg/m^3 .

d: Friction coefficient varies with surface.

QUESTION 56

What rice type is formed when the bran and germ are completely removed?

- a. Brown rice
- b. Well-milled rice
- c. Overmilled rice
- d. Regular-milled rice

ANSWER



c. Overmilled rice

Explanation: Overmilled rice has all bran/germ removed.

a: Only husk is removed.

b: Well-milled retains some streaks.

d: Regular retains up to 40% streaks.

QUESTION 57

Which is a function of an aspiration system in milling?

- a. Shell corn
- b. Dry grains
- c. Remove husks
- d. Grade kernels

ANSWER



c. Remove husks

Explanation: Aspirators blow off light husks.

a: Shellers are for corn.

b: Dryers are separate.

d: Grading is by size.

QUESTION 58

In friction whitening, bran is removed by:

- a. Air jets
- b. Friction between grains
- c. Abrasive surface
- d. Stone peeler

ANSWER



b. Friction between grains

Explanation: It uses grain-to-grain rubbing.

a: Not in friction type.

c: Abrasive surface is for abrasive whitener.

d: No stone involved.

QUESTION 59

What grain component is removed in dehusking?

- a. Bran
- b. Endosperm
- c. Husk
- d. Germ

ANSWER



c. Husk

Explanation: Dehusking removes the outer husk.

a, d: Bran/germ are removed later.

b: Endosperm is retained.

QUESTION 60

Which of the following has the highest milling recovery among dehulling systems?

- a. Kiskisan
- b. Rubber roll
- c. Cono-type
- d. Stone grinder

ANSWER



b. Rubber roll

Explanation: Rubber roll achieves ~70% recovery.

a: Kiskisan yields 50–60%.

c: Cono is lower than rubber roll.

d: Not a standard system for rice.

QUESTION 61

Which material has the highest coefficient of friction with paddy?

- a. Smooth steel
- b. Concrete
- c. Smooth wood
- d. Glass

ANSWER



b. Concrete

Explanation: According to data, concrete = 0.52, higher than others.

a: 0.41

c: 0.44

d: Not in reference.

QUESTION 62

What term refers to grains distinctly damaged by insects or fungi?

- a. Broken
- b. Chalky
- c. Damaged grain
- d. Mottled

ANSWER



c. Damaged grain

Explanation: Includes insect, water, fungal damage.

a: Broken refers to size.

b: Chalky is internal brittleness.

d: Not a standard term.

QUESTION 63

The small broken rice passing through a 1.4 mm sieve is called:

- a. Head rice
- b. Brewer's rice
- c. Small broken
- d. Chalky rice

ANSWER



b. Brewer's rice

Explanation: These are the smallest breakages.

a: Full-sized grains.

c: Slightly larger than brewers.

d: Visual defect, not size class.

QUESTION 64

A pilot light that goes off after ignition is classified as:

- a. Continuous
- b. Standing
- c. Interruptible
- d. Indirect

ANSWER



c. Interruptible

Explanation: It turns off after combustion is established.

a: Stays on all the time.

b: Manually lit and continuous.

d: Refers to dryer type, not pilot.

QUESTION 65

What term defines the movement of air through product during drying?

- a. Aeration
- b. Ventilation
- c. Airflow
- d. Drying front

ANSWER



c. Airflow

Explanation: Airflow represents actual air movement.

a: Low-rate air for storage.

b: General room air movement.

d: A layer within drying grain.

QUESTION 66

What is the purpose of grading in rice milling?

- a. Remove foreign matter
- b. Separate husk
- c. Remove germ
- d. Classify broken and unbroken grains

ANSWER



d. Classify broken and unbroken grains

Explanation: Grading identifies size fractions.

a, b, c: Done in earlier operations.

QUESTION 67

What do we call stored corn grains of the flint variety?

- a. Shelled corn
- b. Corn grits
- c. White corn
- d. Sweet corn

ANSWER



a. Shelled corn

Explanation: Flint/dent corn post-threshing = shelled corn.

b: Grits are milled form.

c & d: Not process terms.

QUESTION 68

Which of the following is NOT a reason for using aeration?

- a. Cool the grain
- b. Prevent odor
- c. Increase grain moisture
- d. Apply fumigants

ANSWER



c. Increase grain moisture

Explanation: Aeration reduces or maintains moisture.

a, b, d: True purposes of aeration.

QUESTION 69

Which term refers to drying where the product remains stationary?

- a. Continuous-flow
- b. Batch dryer
- c. Tunnel dryer
- d. Fluidized bed dryer

ANSWER



b. Batch dryer

Explanation: Batch systems process static grain loads.

a: Grain is moving.

c, d: May involve movement or suspension.

QUESTION 70

What is the primary advantage of bulk storage over bag storage?

- a. Cheaper
- b. Labor intensive
- c. Easy to monitor and fumigate
- d. Flexible storage design

ANSWER



c. Easy to monitor and fumigate

Explanation: Bulk allows easier pest and moisture control.

a: Has higher capital cost.

b: Bag storage is more labor intensive.

d: Bags are more flexible.

QUESTION 71

Which of the following refers to the ratio of weight of vapor in air to its maximum capacity at that temperature?

- a. Relative humidity
- b. Absolute humidity
- c. Equilibrium moisture
- d. Moisture content

ANSWER



a. Relative humidity

Explanation: It is a percentage of saturation.

b: Absolute is actual water content.

c: Refers to product-air balance.

d: Moisture in product, not air.

QUESTION 72

What process prepares corn for final milling by removing hull and germ?

- a. Dehusking
- b. Gritting
- c. Shelling
- d. Conditioning

ANSWER



b. Gritting

Explanation: Grits are produced by removing hull and germ.

a: Rice operation.

c: Only removes kernel from cob.

d: Moisture adjustment step.

QUESTION 73

What dryer type mixes or agitates grain during drying?

- a. Tunnel dryer
- b. Circulating batch dryer
- c. In-storage dryer
- d. Fluidized dryer

ANSWER



b. Circulating batch dryer

Explanation: Designed to mix grain for uniform drying.

a: May not mix grain.

c: Doesn't circulate grain.

d: Suspends grain but doesn't mix batch loads.

QUESTION 74

What type of separator utilizes width and thickness difference between paddy and brown rice?

- a. Tray-type
- b. Seven-type
- c. Compartment-type
- d. Aspirator

ANSWER



b. Seven-type

Explanation: It uses thickness and width for separation.

a: Uses gravity and length.

c: Uses buoyancy.

d: Removes husk.

QUESTION 75

Which component measures drying air pressure inside a system?

- a. CFM meter
- b. Thermocouple
- c. Static pressure gauge
- d. Regulator

ANSWER



c. Static pressure gauge

Explanation: Measures airflow resistance (pressure).

a: Airflow rate only.

b: Measures temperature.

d: Controls fuel or airflow, not measures.

QUESTION 76

Which of the following is used to rapidly remove moisture from crops to a very low level?

- a. Dehumidification
- b. Dehydration
- c. Ventilation
- d. Evaporation

ANSWER



b. Dehydration

Explanation: Dehydration refers to rapid and significant moisture removal.

a: Dehumidification refers to air drying, not crops.

c: Ventilation refers to general air movement.

d: Evaporation is the process, not the drying method.

QUESTION 77

What component is responsible for modulating fuel flow in response to temperature during continuous flow drying?

- a. Valve
- b. Modulator
- c. Pressure regulator
- d. Temperature-sensitive control

ANSWER



d. Temperature-sensitive control

Explanation: It automatically adjusts fuel based on air temp.

a: Valves open/close flow, not modulate.

b: "Modulator" is not the actual sensor.

c: Maintains pressure, not temperature.

QUESTION 78

Which device opens to release excess fuel pressure?

- a. Solenoid valve
- b. Pressure relief valve
- c. Orifice
- d. Burner nozzle

ANSWER



b. Pressure relief valve

Explanation: Opens when pressure exceeds set point.

a: Electrically controlled valve.

c: Just an opening.

d: Delivers fuel, not pressure-regulating.

QUESTION 79

What is the standard term for the warehouse used to store paddy or rice?

- a. Silage bin
- b. Godown
- c. Granary
- d. Hopper

ANSWER



b. Godown

Explanation: Commonly used term in Asia for such warehouses.

a: For ensiling fodder.

c: Old term, rarely used now.

d: Hopper is a funnel-shaped container.

QUESTION 80

Which grain defect results from heating or fermentation that turns kernels yellowish?

- a. Yellow grains
- b. Mottled grains
- c. Chalky grains
- d. Overmilled grains

ANSWER



a. Yellow grains

Explanation: Yellow coloration is linked to damage and fermentation.

b: Not standard terminology here.

c: Chalky grains are brittle and white, not yellow.

d: Overmilled is a process, not a defect.

QUESTION 81

What is the minimum length of grain to be classified as head rice?

- a. $3/8$
- b. $6/8$
- c. $5/8$
- d. $4/8$

ANSWER



b. 6/8

Explanation: Grain $\geq 6/8$ of the original length is head rice.

a, c, d: Represent large or small broken classifications.

QUESTION 82

What is the correct term for a mixture of different grain sizes being sorted to uniform length?

- a. Polishing
- b. Grading
- c. Cleaning
- d. Conditioning

ANSWER



b. Grading

Explanation: Grading separates grain by size.

a: Appearance enhancement process.

c: Removes foreign matter.

d: Adjusts moisture/temp.

QUESTION 83

Which component in the aeration system ensures uniform distribution of air from fan to grain bulk?

- a. Worm
- b. Hopper
- c. Duct
- d. Separator

ANSWER



c. Duct

Explanation: Ducts distribute air from the fan.

a: For feeding grain in milling.

b: Holds grain, doesn't distribute air.

d: Separates grains, not airflow.

QUESTION 84

Which variety of palay is categorized based on aroma and special eating quality?

- a. Short grain
- b. Fancy variety
- c. Traditional variety
- d. Modern variety

ANSWER



b. Fancy variety

Explanation: Fancy varieties include Dinorado, Milagrosa, etc.

a: Size-based.

c: Refers to old, indigenous varieties.

d: Bred for yield, not flavor.

QUESTION 85

Which rice defect shows a significant amount of white opaque surface on the kernel?

- a. Damaged grain
- b. Red-streaked grain
- c. Chalky grain
- d. Polished grain

ANSWER



c. Chalky grain

Explanation: Has a chalky, brittle appearance.

a: From insects or water.

b: Has red streaks.

d: Polished grain is clean and shiny.

QUESTION 86

Which refers to the grain that is less than $3/8$ the length of whole kernel but not passing 1.4 mm sieve?

- a. Brewer's rice
- b. Small broken
- c. Large broken
- d. Head rice

ANSWER



b. Small broken

Explanation: Small broken are $<3/8$ and retained on sieve.

a: Must pass sieve.

c: $3/8$ to $<6/8$.

d: $\geq 6/8$.

QUESTION 87

What component of a drying system ignites and maintains combustion?

- a. Pilot
- b. Burner
- c. Orifice
- d. Vaporizer

ANSWER



a. Pilot

Explanation: Pilot ignites the burner flame.

b: Actual combustion source.

c: Opening for fuel flow.

d: Converts fuel from liquid to gas.

QUESTION 88

Which type of control shuts down a dryer if any component malfunctions?

- a. Thermostat
- b. Manual override
- c. Fail-safe control
- d. Modulator

ANSWER



c. Fail-safe control

Explanation: It ensures automatic shutdown during abnormal operation.

a: Controls temp only.

b: Manual, not automatic.

d: Regulates fuel based on temp.

QUESTION 89

What postharvest tool takes samples from bagged or bulk grains?

- a. Trier
- b. Probe
- c. Sifter
- d. Planter

ANSWER



a. Trier

Explanation: Trier is a metal probe used in grain sampling.

b: Generic; not specific.

c: Used for sorting grains by size.

d: Planting equipment.

QUESTION 90

What drying term refers to the moving layer between dried and undried material?

- a. Drying line
- b. Evaporation boundary
- c. Drying front
- d. Heat wave

ANSWER



c. Drying front

Explanation: This is where drying is actively occurring.

a, b, d: Not standard terms.

QUESTION 91

Which of the following is a standard unit for fuel energy content?

- a. PSI
- b. BTU
- c. Pascal
- d. Joule

ANSWER



d. Joule

Explanation: Joule is SI unit of heat.

b: British unit, also acceptable.

a, c: Units of pressure.

QUESTION 92

What is the term for the combination of radiant energy emission, transmission, and absorption?

- a. Induction
- b. Conduction
- c. Radiation
- d. Convection

ANSWER



c. Radiation

Explanation: Radiation covers all three processes.

a: Related to electricity.

b: Heat transfer via contact.

d: Heat via moving fluid.

QUESTION 93

What is the name for a manually lit pilot that stays on until manually shut off?

- a. Interruptible
- b. Continuous
- c. Standing
- d. Auto-light

ANSWER



c. Standing

Explanation: It must be lit and shut off manually.

a: Turns off after ignition.

b: Stays on but may be auto-lit.

d: Not a defined category here.

QUESTION 94

What type of grain storage is flexible but labor-intensive and prone to spillage?

- a. Silo storage
- b. Bulk storage
- c. Bag storage
- d. Drum storage

ANSWER



c. Bag storage

Explanation: Bags are easy to manage but laborious.

a, b: Require mechanical handling.

d: Not standard in this context.

QUESTION 95

What term refers to the measure of heat added by the burner to air, shown by increased temp?

- a. Temperature rise
- b. Drying gain
- c. Heat index
- d. Fuel utilization

ANSWER



a. Temperature rise

Explanation: Difference between ambient and drying air temp.

b, c, d: Not technical drying terms.

QUESTION 96

Which of the following refers to the speed at which air moves through the product void space?

- a. Apparent velocity
- b. Cross velocity
- c. Average velocity
- d. Linear velocity

ANSWER



c. Average velocity

Explanation: It's based on actual flow through pores.

a: Calculated without adjusting for voids.

b, d: Not used in this context.

QUESTION 97

A type of dryer where grain is carried through a tunnel-like chamber:

- a. Batch dryer
- b. Tunnel dryer
- c. Counter-flow dryer
- d. Rotary dryer

ANSWER



b. Tunnel dryer

Explanation: Grain is conveyed through a tunnel.

a: Stationary batch drying.

c: Air and grain flow in opposite directions.

d: Not mentioned in reference.

QUESTION 98

A drying method where combustion gases never touch the grain:

- a. Direct-fired
- b. Indirect-fired
- c. Ventilated
- d. Radiant

ANSWER



b. Indirect-fired

Explanation: Heat is transferred through surfaces.

a: Gases contact grain.

c, d: Not defined as dryer types.

QUESTION 99

A safety device that automatically shuts off gas when flow exceeds the set limit:

- a. Orifice
- b. Pilot valve
- c. Excess-flow valve
- d. Relief valve

ANSWER



c. Excess-flow valve

Explanation: Closes during abnormal flow conditions.

a: Simple opening.

b: Controls ignition.

d: Opens when pressure is high.

QUESTION 100

What drying metric represents the ratio of cooling to heating in air during drying?

- a. Heat loss factor
- b. Cooling index
- c. Heat utilization factor (HUF)
- d. Temperature gradient

ANSWER



c. Heat utilization factor (HUF)

Explanation: $HUF = (T1 - T3) / (T2 - T1)$, ratio of evaporative cooling to heating.

a, b, d: Incorrect relationship

PART II.

Computational Multiple-Choice Questions

QUESTION 1

A 100 g sample of freshly harvested corn contains 28 g of water. What is the moisture content on a wet basis (wb)?

- a. 22.0%
- b. 28.0%
- c. 38.9%
- d. 72.0%

ANSWER



b. 28.0%

$$\begin{aligned} MC_{wb} &= (\text{Weight of water} / \text{Total weight}) \times 100 \\ &= (28 / 100) \times 100 = 28.0\% \end{aligned}$$

QUESTION 2

A 100 g sample of grain has 25 g of water. What is the moisture content on a dry basis (db)?

- a. 25.0%
- b. 33.3%
- c. 75.0%
- d. 20.0%

ANSWER



b. 33.3%

$$\text{MCdb} = (\text{Weight of water} / \text{Dry matter}) \times 100$$

$$\text{Dry matter} = 100 - 25 = 75$$

$$\text{MCdb} = (25 / 75) \times 100 = 33.3\%$$

QUESTION 3

Convert 40% moisture content dry basis to wet basis.

- a. 28.6%
- b. 35.0%
- c. 40.0%
- d. 60.0%

ANSWER



a. 28.6%

$$\begin{aligned} MC_{wb} &= [MC_{db} / (100 + MC_{db})] \times 100 \\ &= [40 / (100 + 40)] \times 100 = (40/140) \times 100 = 28.57\% \end{aligned}$$

QUESTION 4

Convert 20% wet basis moisture content to dry basis.

- a. 25.0%
- b. 16.7%
- c. 20.0%
- d. 18.5%

ANSWER



a. 25.0%

$$\begin{aligned} MC_{db} &= [MC_{wb} / (100 - MC_{wb})] \times 100 \\ &= [20 / (100 - 20)] \times 100 = 25.0\% \end{aligned}$$

QUESTION 5

A corn sample weighs 120 g before drying and 96 g after drying. What is its initial moisture content (wb)?

- a. 20.0%
- b. 25.0%
- c. 40.0%
- d. 33.3%

ANSWER



b. 20.0%

Water loss = $120 - 96 = 24 \text{ g}$

$\text{MC}_{\text{wb}} = (24 / 120) \times 100 = 20.0\%$

QUESTION 6

A batch of rice has 15% MC_{wb}. If the batch weighs 200 kg, what is the weight of dry matter?

- a. 170 kg
- b. 180 kg
- c. 150 kg
- d. 190 kg

ANSWER



b. 170 kg

$$\begin{aligned}\text{Dry weight} &= \text{Total weight} \times (1 - MC_{wb}) \\ &= 200 \times (1 - 0.15) = 200 \times 0.85 = 170 \text{ kg}\end{aligned}$$

QUESTION 7

What is the moisture loss if 1,000 kg of corn at 25% MC_{wb} is dried to 14% MC_{wb}?

- a. 110.5 kg
- b. 127.9 kg
- c. 140.0 kg
- d. 116.3 kg

ANSWER



b. 127.9 kg

Dry matter = $1000 \times (1 - 0.25) = 750 \text{ kg}$

Final weight at 14% MC_{wb} = $750 / 0.86 = 872.1 \text{ kg}$

Moisture loss = $1000 - 872.1 = 127.9 \text{ kg}$

QUESTION 8

A 500 kg batch of rice at 18% MC_{wb} is dried to 14% MC_{wb}. What is the final weight?

- a. 430.0 kg
- b. 450.0 kg
- c. 465.1 kg
- d. 476.7 kg

ANSWER



d. 476.7 kg

Dry matter = $500 \times (1 - 0.18) = 410 \text{ kg}$

Final weight = $410 / 0.86 = 476.7 \text{ kg}$

QUESTION 9

Which formula converts MC from wet basis to dry basis?

- a. $(MC_{wb} / (100 - MC_{wb})) \times 100$
- b. $(100 - MC_{wb}) / MC_{wb} \times 100$
- c. $(MC_{wb} \times 100) / (100 + MC_{wb})$
- d. $(MC_{wb} - 100) \times 100$

ANSWER



a. $(MC_{wb} / (100 - MC_{wb})) \times 100$

c: Used for db to wb

QUESTION 10

You have a 10 kg corn sample at 28% MC_{wb}. How much water must be removed to bring it to 14% MC_{wb}?

- a. 1.5 kg
- b. 2.0 kg
- c. 1.6 kg
- d. 2.6 kg

ANSWER



c. 1.6 kg

Dry matter = $10 \times 0.72 = 7.2$ kg

Final weight = $7.2 / 0.86 = 8.37$ kg

Water to remove = $10 - 8.37 = 1.63$ kg

QUESTION 11

A batch dryer dries 1,000 kg of paddy in 6 hours.
What is the average drying rate?

- a. 150 kg/hr
- b. 167 kg/hr
- c. 125 kg/hr
- d. 200 kg/hr

ANSWER



b. 167 kg/hr

Drying rate = Total weight / Time = $1000 / 6 \approx 166.7$ kg/hr

QUESTION 12

A dryer processes 600 kg of corn in 4 hours. What is the dryer capacity?

- a. 150 kg/hr
- b. 120 kg/hr
- c. 100 kg/hr
- d. 175 kg/hr

ANSWER



a. 150 kg/hr

Capacity = $600 / 4 = 150$ kg/hr

QUESTION 13

A grain dryer uses 3 liters of diesel per hour. If it operates for 5 hours, how much fuel is used?

- a. 10 L
- b. 12 L
- c. 15 L
- d. 18 L

ANSWER



c. 15 L

Fuel = $3 \times 5 = 15$ liters

QUESTION 14

If a dryer consumes 10 L of diesel costing ₱55 per liter, what is the fuel cost for one drying cycle?

- a. ₱500
- b. ₱550
- c. ₱600
- d. ₱650

ANSWER



b. ₱550

$$\text{Cost} = 10 \times 55 = 550$$

QUESTION 15

A rice dryer reduces moisture from 22% to 14%. If initial weight is 1,000 kg, what is final weight?

- a. 920.0 kg
- b. 880.0 kg
- c. 860.0 kg
- d. 906.9 kg

ANSWER



d. 906.9 kg

Dry matter = $1000 \times 0.78 = 780$

Final weight = $780 / 0.86 \approx 906.9 \text{ kg}$

QUESTION 16

A drying operation removes 120 kg of moisture in 8 hours. What is the average drying rate?

- a. 10 kg/hr
- b. 15 kg/hr
- c. 20 kg/hr
- d. 12 kg/hr

ANSWER



b. 15 kg/hr

Drying rate = $120 / 8 = 15 \text{ kg/hr}$

QUESTION 17

A 400-kg batch is dried from 24% to 14%. How much water was removed?

- a. 46.5 kg
- b. 56.5 kg
- c. 65.5 kg
- d. 85.5 kg

ANSWER



a. 46.5 kg

Dry matter = $400 \times 0.76 = 304$ kg

Final weight = $304 / 0.86 \approx 353.5$ kg

Water removed = $400 - 353.5 = \mathbf{46.5 \text{ kg}}$

QUESTION 18

A dryer can dry 800 kg of grain in 5 hours. How long will it take to dry 1,200 kg?

- a. 6.5 hrs
- b. 7.0 hrs
- c. 7.5 hrs
- d. 8.0 hrs

ANSWER



c. 7.5 hrs

Rate = $800 / 5 = 160$ kg/hr

Time = $1200 / 160 = 7.5$ hrs

QUESTION 19

How much moisture is in 250 kg of rice at 18% MC_{wb} ?

- a. 45.0 kg
- b. 30.0 kg
- c. 37.5 kg
- d. 50.0 kg

ANSWER



a. 45.0 kg

$$\text{Water} = 250 \times 0.18 = 45 \text{ kg}$$

QUESTION 20

What is the final moisture content if a 100 kg corn sample loses 18 kg of water?

- a. 18%
- b. 82%
- c. 22%
- d. 32%

ANSWER



c. 22%

Final weight = $100 - 18 = 82$ kg

$MC_{wb} = 18 / 82 \times 100 \approx 22\%$

QUESTION 21

A dryer reduces a 1,200 kg batch of paddy from 24% to 14% MC_{wb} in 6 hours. What is the average drying rate in kg of water removed per hour?

- a. 20.0 kg/hr
- b. 25.0 kg/hr
- c. 30.0 kg/hr
- d. 35.0 kg/hr

ANSWER



a. 20.0 kg/hr

$$\begin{aligned}\text{MC removed} &= (\text{Initial MC} - \text{Final MC}) \times \text{Initial Weight} \\ &= (0.24 - 0.14) \times 1200 = 120 \text{ kg}\end{aligned}$$

$$\text{Drying rate} = 120 / 6 = 20.0 \text{ kg/hr}$$

QUESTION 22

A 1,500 kg batch of corn is dried from 26% to 14% MC_{wb} using a dryer with 90% efficiency. If each liter of fuel provides 35 MJ and the latent heat of vaporization of water is 2,257 kJ/kg, estimate the liters of fuel required.

- a. 13.0 L
- b. 14.5 L
- c. 16.0 L
- d. 17.0 L

ANSWER



a. 13.0 L

Water removed = $(0.26 - 0.14) \times 1500 = 180 \text{ kg}$

Energy needed = $180 \times 2257 = 406,260 \text{ kJ}$

Energy input = $406,260 / 0.90 = 451,400 \text{ kJ}$

MJ = $451,400 / 1000 = 451.4 \text{ MJ}$

Fuel needed = $451.4 / 35 \approx 12.9 \text{ L}$

QUESTION 23

A dryer can handle 3 layers of rice trays, each drying 100 kg from 22% to 14% MC_{wb} in 5 hours. If the system adds a 4th tray, but drying time extends by 15%, what is the new drying time?

- a. 5.5 hrs
- b. 5.75 hrs
- c. 6.0 hrs
- d. 6.25 hrs

ANSWER



b. 5.75 hrs

Increase = 15% of 5 hrs = 0.75 hr

New time = 5 + 0.75 = 5.75 hrs

QUESTION 24

A rewetting operation raises moisture from 10% to 14% MC_{wb} by adding water. How much water must be added to 500 kg of dry grain?

- a. 23.3 kg
- b. 26.7 kg
- c. 32.6 kg
- d. 35.4 kg

ANSWER



a. 23.3 kg

Final weight (with 14% MC) = Dry matter / (1 - 0.14)

Dry = 500 kg

Final total = 500 / 0.86 = 581.4 kg

Water added = 581.4 - 500 = 81.4 kg

Dry grain is not **500 kg** if **MCwb is 10%**

Dry matter = 500 × 0.90 = 450 kg

Final weight = 450 / 0.86 = 523.3 kg

Water to add = 523.3 - 500 = 23.3 kg

QUESTION 25

Which of the following is not a reason why grain is tempered during drying?

- a. To reduce drying time
- b. To allow moisture redistribution
- c. To improve dryer fuel efficiency
- d. To improve product color

ANSWER



d. To improve product color

Explanation: Tempering equalizes internal moisture, not for color.

a, b, c: valid effects of tempering

QUESTION 26

A continuous-flow dryer has a throughput of 1,000 kg/hr and moisture reduction of 10% (wb). How much water is removed per hour?

- a. 100 kg
- b. 150 kg
- c. 200 kg
- d. 250 kg

ANSWER



a. 100 kg

Water removed = 10% of 1000 = 100 kg/hr

QUESTION 27

A batch of paddy is 24% MC_{wb} and weighs 1,200 kg. What will be the weight after drying to 14% MC_{wb}?

- a. 1,070 kg
- b. 1,065 kg
- c. 1,060 kg
- d. 1,050 kg

ANSWER



b. 1,065 kg

Dry matter = $1200 \times 0.76 = 912$

Final weight = $912 / 0.86 = 1,060.5 \text{ kg}$

QUESTION 28

Which equation is used to estimate drying time?

- a. $t = \Delta M / R$
- b. $t = R / \Delta M$
- c. $t = R \times \Delta M$
- d. $t = \Delta M^2 / R$

ANSWER



$$\mathbf{a. \text{ } t = \Delta M / R}$$

Drying time = moisture to be removed / drying rate

QUESTION 29

A drying fan consumes 1.5 kW of power. If used for 6 hours, what is the energy consumption in kWh?

- a. 6.0 kWh
- b. 7.5 kWh
- c. 9.0 kWh
- d. 10.5 kWh

ANSWER



c. 9.0 kWh

$$\text{Energy} = 1.5 \times 6 = 9.0 \text{ kWh}$$

QUESTION 30

Which of the following combinations will result in the longest drying time?

- a. High air temp, low airflow
- b. Low air temp, high humidity
- c. Medium air temp, low humidity
- d. High airflow, high temperature

ANSWER



b. Low air temp, high humidity

Explanation: Low drying potential due to low heat and poor moisture gradient.

a, c, d: conditions with improved drying efficiency

QUESTION 31

The weight of newly dried paddy rice is 1,299.6 kg. If the moisture content is 14% dry basis, what is the amount of dry matter in kg?

- a. 1,140 kg
- b. 1,200 kg
- c. 159.6 kg
- d. 200.4 kg

ANSWER



a. 1,140 kg

$$\text{Dry Matter} = \frac{W}{1 + \frac{MC_{db}}{100}}$$

W = total weight (newly dried paddy rice) = 1,299.6 kg

MC_{db} = 14%

$$\text{Dry Matter} = \frac{1,299.6}{1 + \frac{14}{100}} = \frac{1,299.6}{1.14} \approx 1,140.0 \text{ kg}$$

QUESTION 32

What is the remaining moisture, in kg, in item No. 31?

- a. 1,140 kg
- b. 159.6 kg
- c. 200.4 kg
- d. 140 kg

ANSWER



b. 159.6 kg

Total weight = 1,299.6 kg

Dry matter = 1,140.0 kg

Moisture = Total weight – Dry matter
= 1,299.6 – 1,140.0 = 159.6kg

QUESTION 33

What is the moisture loss in the paddy rice in item No.31 if the initial weight (weight before drying) was 1,500 kg?

- a. 1,140 kg
- b. 159.6 kg
- c. 200.4 kg
- d. 140 kg

ANSWER



c. 200.4 kg

Initial weight (before drying): 1,500 kg

Final weight (after drying): 1,299.6 kg

Moisture loss = Initial weight – Final weight

Moisture loss = 1,500 – 1,299.6 = 200.4kg

QUESTION 34

What was the moisture content wet basis of the paddy rice in item No.31 if the initial weight was 1,500 kg?

- a. 14%
- b. 18%
- c. 22%
- d. 24%

ANSWER



d. 24%

Initial weight (before drying): 1,500 kg

Dry matter (from previous item): 1,140.0 kg

$$MC_{wb} = \frac{\text{Water weight}}{\text{Total weight}} \times 100$$

$$\text{Water weight} = 1,500 - 1,140 = 360 \text{ kg}$$

$$MC_{wb} = \frac{360}{1,500} \times 100 = 24\%$$

QUESTION 35

What is the equivalent dry basis moisture content in item No. 34?

- a. 36.2%
- b. 31.6%
- c. 28.3%
- d. 22.4%

ANSWER



b. 31.6%

$$MC_{db} = \frac{MC_{wb}}{1 - MC_{wb}/100} \times 100$$

$$MC_{db} = \frac{24}{1 - 0.24} \times 100 = \frac{24}{0.76} \times 100 \approx 31.58\%$$



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THANK YOU

